Aircraft Optimal Design

Aerospace Engineering

Prof. André C. Marta

Afonso Vieira, Cândida Miranda, Gonçalo Teixeira, Guilherme Rodrigues

116488, 106600, 116577, 119211

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1 Introduction

This report documents a conceptual aerostructural design study of a regional turboprop aircraft, carried out with the open-source framework `OpenAeroStruct` (OAS) [6] built on top of the `OpenMDAO` multidisciplinary optimization platform [8]. The ATR 72-600 was chosen as the reference aircraft, and its main wing and horizontal tail are sized simultaneously by a gradient-based optimizer that minimizes cruise fuel burn subject to trim and structural constraints.

1.1 Motivation

In conventional sequential design, the aerodynamicist and the structural engineer optimize their disciplines in turn, which ignores the strong two-way coupling between the aerodynamic load and the elastic deformation of the wing. A lighter wing bends more, which changes the spanwise lift distribution, which in turn changes the load that sizes the structure. Multidisciplinary Design Optimization (MDO) resolves this coupling by treating the aircraft as a single coupled system [4]. For a fuel-sensitive regional turboprop such as the ATR 72, even a few percent reduction in cruise fuel burn translates directly into lower operating cost and emissions, which makes the aerostructural trade-off worth solving consistently rather than discipline by discipline.

1.2 Topic Overview

`OpenAeroStruct` couples a Vortex-Lattice Method (VLM) for the aerodynamics with a 6-degree-of-freedom spatial-beam (wingbox) Finite Element Method (FEM) for the structure. The two disciplines are solved as a fixed-point aeroelastic problem, and analytic (adjoint) derivatives provided by `OpenMDAO` make gradient-based optimization with dozens of design variables affordable. The present study uses a two-point (multipoint) formulation: a $1g$ cruise point at FL250, which defines the objective and the trim constraints, and a $2.5g$ symmetric pull-up point at sea level, which sizes the wing and tail spar/skin against material failure as required by EASA CS-25 [3].

1.3 Objectives

The objectives of this work are to:

- Define the Baseline and Flight Conditions: Establish the reference aircraft parameters, structural material properties, and the operational flight envelope to serve as the foundation of the study (Section 2).
- Implement the Aerostructural Coupling: Set up the individual aerodynamic and structural computational models, and develop the multidisciplinary coupling framework, represented via an N2 diagram, to enable accurate Multidisciplinary Design Analysis (MDA) (Sections 3.1 and 3.2).
- Verify Numerical Accuracy: Conduct grid convergence and solver tolerance studies to guarantee the numerical verification, reliability, and efficiency of the computational framework (Section 3.4).

- Formulate the MDO Problem: Rigorously formulate the optimization problem by defining the design variables, geometric and structural constraints, and the multi-point flight configurations (Section 4).
- Analyze the Optimization Results: Execute the optimization process, evaluate the resulting optimal aerodynamic shapes and structural layouts, and discuss the performance improvements and trade-offs against the baseline aircraft (Section 5).

2 Problem Statement

The reference aircraft is the ATR 72-600, a twin-turboprop regional airliner. It was chosen because it cruises slowly ($M \approx 0.44$), so the incompressible VLM aerodynamics of `OpenAeroStruct` apply with minimal correction, and because its conventional configuration (high wing, almost unswept, plus horizontal tail) maps directly onto the assignment requirements. The parameters were collected from the manufacturer factsheet [2] and the HAW Hamburg design study by Nita [7]. The Nita study addresses the ATR 72 generically (it predates the -600 variant); since the airframe and wing are common to the -500/-600 variants, its wing geometry applies directly to the -600, while the variant-specific data (weights, engine, range) are taken from the ATR 72-600 factsheet. A complete list of input values with the source of each one is given in Appendix A.1. The principal dimensions of the aircraft are reproduced in Figure 1.

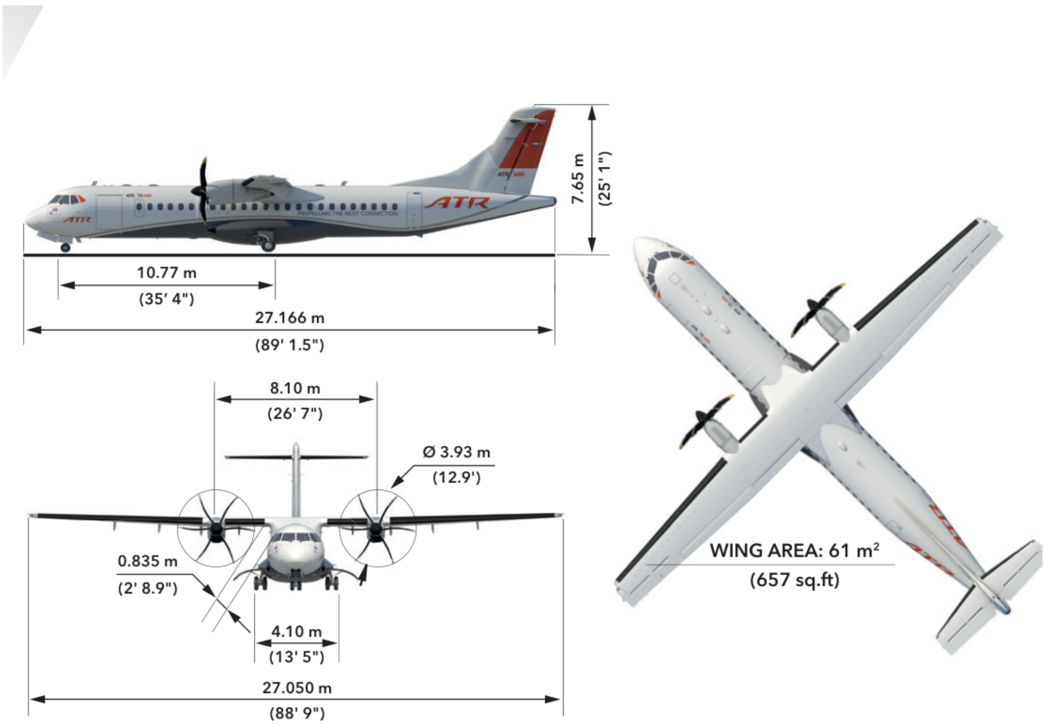


Figure 1: Dimensioned three-view of the ATR 72-600, showing the 27.05 m span, 27.17 m length and 61 m^2 wing area (manufacturer factsheet [2]).

2.1 Baseline Aircraft Parameters

The top-level specifications of the aircraft and its mission are listed in Table 1.

Parameter	Value
Range (max. payload)	1404 km (758 NM)
Cruise Mach number	0.443
Cruise altitude	7620 m (FL250)
Take-off weight (MTOW)	23 000 kg
Operating empty weight (OEW)	$\sim$13 450 kg
Engine	2$\times$ PW127 (turboprop)
Max. lift coefficient (clean wing)	1.374
Limit maneuvering load factor n	2.5

Table 1: Baseline aircraft and mission parameters of the ATR 72-600.

Table 1 fixes the mission (range, cruise point) and the gross weights that drive the Breguet fuel estimate and the structural sizing. The maneuvering load factor $n = 2.5$ is the limit value prescribed by EASA CS-25 [3] for transport aeroplanes.

2.2 Wing and Tail Parameters

The geometry of the two lifting surfaces is summarized in Table 2. The main wing is modelled as an equivalent simple trapezoid (the real ATR wing is double-tapered, with an outer-panel taper $\lambda = 0.59$ and a kink at $\eta = 0.346$); the chosen $\lambda = 0.45$ reproduces the reference area and a realistic mean aerodynamic chord. The horizontal tail is sized by the tail-volume-coefficient method ($V_{HT} = 1.0$, arm $l_{HT} = 12$ m), with an aspect ratio of 4.6 matching the real ATR tailplane.

Parameter	Main wing	Horizontal tail
Area [m^2]	61.0	12.0
Span [m]	27.05	7.43
Aspect ratio	12.0	4.6
Taper ratio	0.45	0.5
Root / tip chord [m]	3.11 / 1.40	2.16 / 1.08
Mean aerodynamic chord [m]	2.36	–
Sweep @$c/4$ [deg]	3	0
Dihedral [deg]	0	0
Airfoil (root/tip)	NACA 43018/43013	NACA 0009
Thickness/chord (root/tip)	0.18 / 0.13	0.09
Moment arm l_{HT} [m]	–	12.0

Table 2: Geometric parameters of the main wing and horizontal tail.

The resulting VLM/FEM planform mesh is shown in Figure 2, together with the two engine point masses (≈ 480 kg each, PW127).

Figure 2 confirms the high-aspect-ratio ($\mathrm{AR} = 12$), lightly tapered wing and a horizontal tail of about 12 m^2 placed 12 m aft.

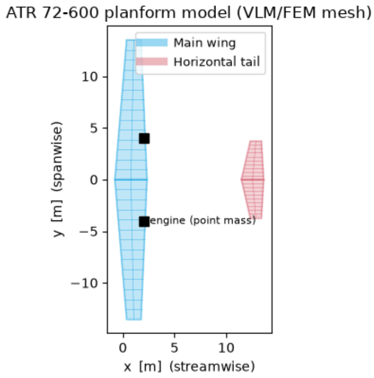


Figure 2: Planform of the wing and horizontal tail as discretized for the VLM/FEM analysis. The black squares mark the engine point masses.

2.3 Material

All primary structure (spar caps and skins of the wingbox) is aluminum 7075-T6, whose properties are listed in Table 3. The yield stress is the bare-alloy value; a safety factor of 1.5 is applied on top.

Property	Value
Young's modulus E	71.7 GPa
Shear modulus G	26.9 GPa
Yield stress σ_y	505 MPa
Density ρ	2810 kg/m^3
Safety factor	1.5

Table 3: Mechanical properties of aluminum 7075-T6.

2.4 Flight Envelope

Two flight conditions are evaluated simultaneously (multipoint): a $1g$ cruise point that defines the objective and the trim constraints, and a $2.5g$ symmetric pull-up that sizes the structure against failure. Their atmospheric conditions are listed in Table 4. The cruise point uses the ISA atmosphere at FL250; the maneuver point is taken at sea level, where the dense air makes the $2.5g$ load reachable below stall (the standard sea-level structural sizing case).

Parameter	Cruise ($1g$)	Maneuver ($2.5g$)
Altitude [m]	7620 (FL250)	0 (sea level)
Load factor	1.0	2.5
Mach number	0.443	0.443
Air density [kg/m^3]	0.549	1.225
Speed of sound [m/s]	309.6	340.3
True airspeed [m/s]	137.2	150.8
Reynolds/chord [1/m]	4.9×10^6	–

Table 4: Flight conditions for the cruise and maneuver points.

The structural sizing load case sits on the maneuver flight envelope (Figure 3), the v-n diagram that bounds the admissible load factor against airspeed. The positive limit $n = +2.5$ and the negative limit $n = -1.0$ follow from EASA CS-25 [3]; the $C_{L,max}$ parabola fixes the stall speed V_S and the corner (maneuver) speed V_A, beyond which the full $2.5g$ can be pulled. The structure is sized at $V_A/2.5g$.

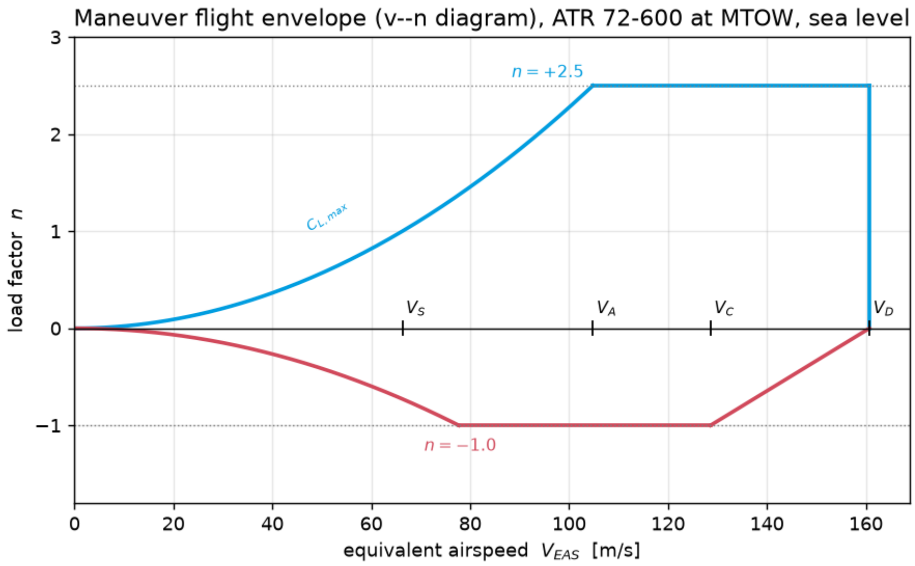


Figure 3: Maneuver flight envelope (v–n diagram) of the ATR 72-600 at MTOW and sea level, with the CS-25 limit load factors $n = +2.5$ and $n = -1.0$ and the characteristic speeds V_S, V_A, V_C and V_D.

Two modelling approximations already appear at this stage: the turboprop fuel consumption is represented through an *equivalent* thrust-specific value (calibrated so that the Breguet fuel reproduces the published ATR cruise consumption, ~760 kg/h), and the double-tapered wing is reduced to an equivalent simple trapezoid. Both are revisited in Section 6.

3 Software Setup

The design is carried out with `OpenAeroStruct` (OAS), an aerostructural optimization framework built on the `OpenMDAO` platform. This section summarizes the physical models, their coupling, the optimizer, and the verification of the numerical settings.

3.1 Aerodynamic and Structural Models

Aerodynamics. The lifting surfaces are modelled with a Vortex-Lattice Method (VLM), a potential-flow panel method valid for attached, incompressible to low-subsonic flow. Each surface is discretized into a grid of horseshoe vortices; enforcing flow tangency at the panel collocation points yields a linear system for the circulation $\mathbf{\Gamma}$, from which the spanwise lift and induced drag follow from the Kutta–Joukowski theorem. Profile drag is added through a flat-plate viscous build-up, and a Prandtl–Glauert correction accounts for compressibility. At the cruise Mach number of the ATR 72 ($M \approx 0.44$) wave drag is negligible and is switched off.

Structure. The wingbox is modelled as a tubular spatial beam discretized with 6-DOF finite elements. The cross-section is a two-cell box bounded by the upper/lower skins and the front/rear spars, whose enclosed area and second moments are computed from the airfoil ordinates (generated from the standard NACA thickness equation [5]). The structure is sized in failure: the axial and shear stresses are aggregated with a smooth Kreisselmeier–Steinhauser (KS) function, so that a single constraint $\mathrm{KS} \leq 0$ guarantees that no element exceeds the material yield stress.

3.2 Discipline Coupling

The aerodynamic and structural disciplines form a two-way coupled (aeroelastic) system: the aerodynamic loads deform the elastic wing, and the deformed shape changes the local angle of attack and therefore the loads. This fixed point is solved at each design iteration by a non-linear block Gauss–Seidel (NLBGS) Multidisciplinary Analysis (MDA), accelerated with Aitken relaxation. The residual form is

$$\mathcal{R}(\boldsymbol{u}) = \begin{bmatrix} \mathcal{A}(\mathbf{\Gamma}, \boldsymbol{d}) \\ \mathcal{S}(\boldsymbol{d}, \boldsymbol{f}(\mathbf{\Gamma})) \end{bmatrix} = \mathbf{0}\,, \tag{1}$$

where $\mathbf{\Gamma}$ are the circulations, $\boldsymbol{d}$ the structural displacements and $\boldsymbol{f}$ the transferred loads.

The N^2 diagram obtained is shown in Figure 4. Reading the matrix, squares above the diagonal denote feed-forward data flow and below the diagonal denote feedback. The execution order follows: `Auto-IVC` and `indeps` supply the design variables; `wing_mesh_param` and `tail_mesh_param` generate the meshes from span and sweep; `wing_geom` and `tail_geom` compute the structural and geometric properties; the `cruise` and `maneuver` points solve the non-linear aerostructural coupling iteratively within their `coupled` subgroups (using the **NL: NLBGS** solver, while **LN: Direct** handles the underlying linear systems for analytic derivatives); and `fuel_vol_check` evaluates the fuel volume constraint. The near-fully upper-triangular pattern confirms a globally feed-forward architecture, with no top-level non-linear iteration required.

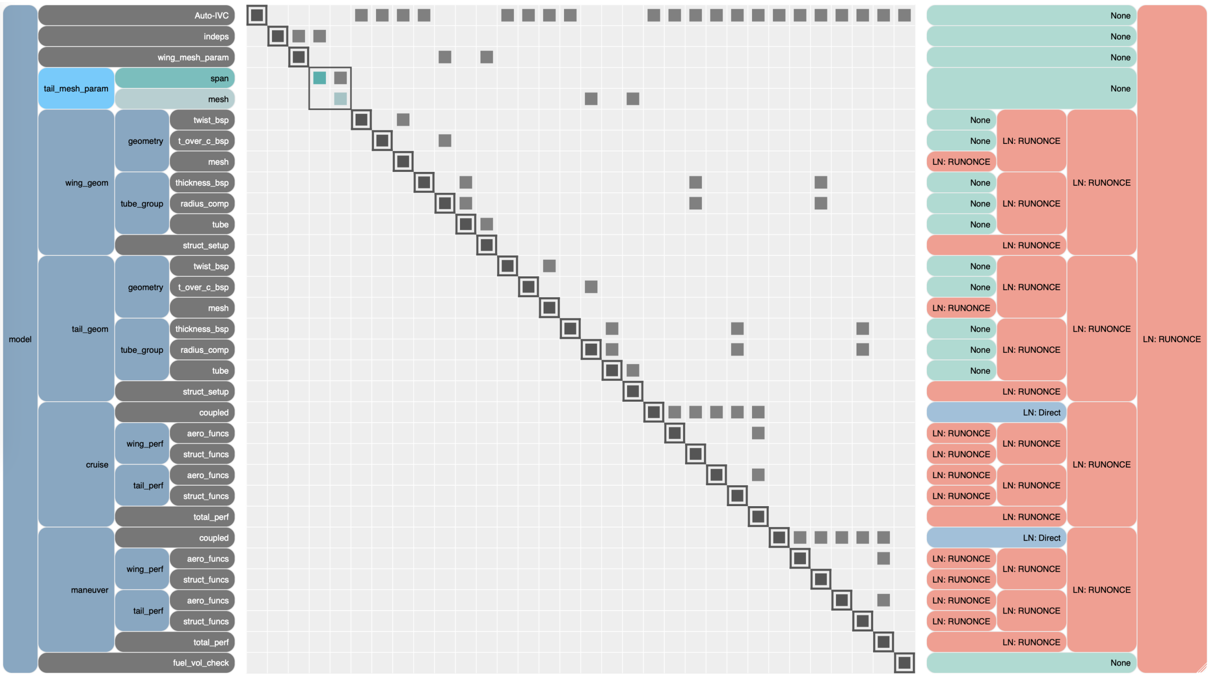


Figure 4: Representation of the model in an N^2 diagram

3.3 Optimizer

The problem is solved with Sequential Least-Squares Quadratic Programming (SLSQP), a gradient-based quasi-Newton method that handles non-linear equality and inequality constraints. Gradient-based methods scale well with the number of design variables, which is essential here because each thickness and twist control point is independent. Their efficiency relies on accurate derivatives: `OpenMDAO` assembles the total derivatives of the coupled system analytically through the Modular Analysis and Unified Derivatives (MAUD) framework, i.e. by solving the coupled adjoint system rather than by finite differences. Being a local method, SLSQP returns a local optimum consistent with the starting point. The optimizer and solver settings are summarized in Table 5.

Setting	Value
Optimizer	SLSQP (gradient-based)
Driver tolerance	10^{-6}
Total derivatives	analytic adjoint (MAUD)
MDA solver	non-linear block Gauss–Seidel (Aitken)
MDA max. iterations	40
Wing mesh (chord $\times$ span)	3×15
Tail mesh (chord $\times$ span)	3×11

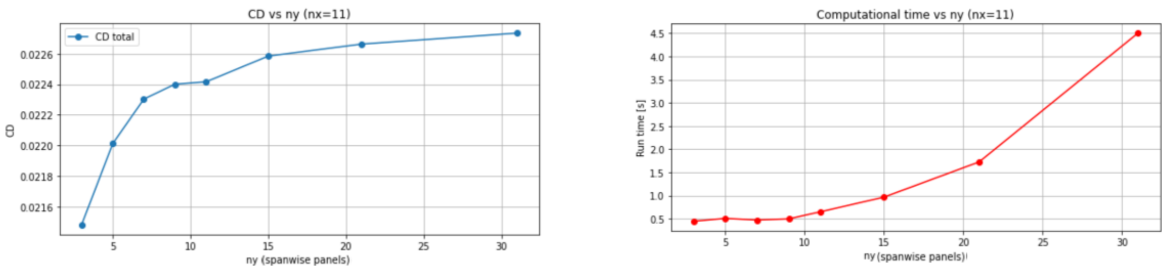
Table 5: Optimizer and aerostructural solver settings used throughout the work.

3.4 Mesh and Tolerance Verification

Mesh convergence.

Spanwise Direction

A mesh refinement study was conducted to ensure that the aerodynamic results are independent of the spatial discretization. Following the conclusions of the mesh study performed in the previous assignment, the chordwise resolution was initially fixed at $n_x = 11$, since this value already provided a converged prediction of the aerodynamic coefficients. With n_x fixed, the present analysis varied the spanwise resolution over the set $n_y \in \{3, 5, 7, 9, 11, 15, 21, 31\}$ in order to quantify the sensitivity of the lift and drag coefficients to the spanwise discretization. The drag coefficient C_D was adopted as the primary convergence metric, and the mesh was considered sufficiently refined when the relative variation in C_D between successive values of n_y fell below 1%. The final mesh resolution was selected as the smallest pair (n_x, n_y) satisfying this convergence criterion while maintaining a reasonable computational cost.



(a) C_D as a function of n_y

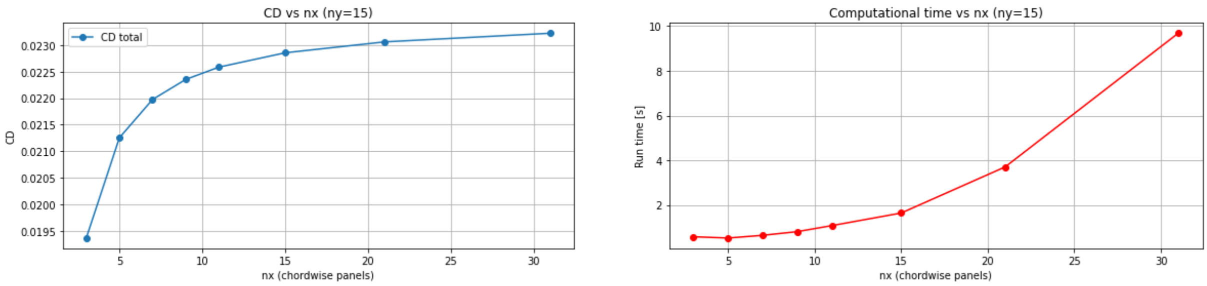
(b) Computational time as a function of n_y

Figure 5: Mesh refinement study in the spanwise direction for fixed $n_x = 11$.

Around $n_y = 15$ the CD value stabilizes while having a relative low computational time $(0.97s)$ so this was the chosen value for n_y for the next mesh study, n_x.

Chordwise Direction

For fixed spanwise resolution $n_y = 15$, the drag coefficient increases slightly with the number of chordwise panels n_x. This behaviour is expected in OpenAeroStruct, since chordwise refinement improves the geometric representation of the camber and structural deformation, leading to small but consistent changes in the induced drag. The resulting curve is smooth and monotonic, and it stabilises around $n_x = 15$. However, applying the 1% convergence criterion to C_D shows that the solution only becomes fully mesh-independent at $n_x = 21$, which is therefore the smallest chordwise resolution satisfying the convergence requirement.



(a) C_D as a function of n_x

(b) Computational time as a function of n_x

Figure 6: Mesh refinement study in the chordwise direction for fixed $n_y = 15$.

Even though the mesh study indicated that the optimal configuration was $n_y = 15$ and $n_x = 21$, this resolution could not be used in the advanced optimization runs. At such high discretization levels, the computational cost became prohibitive and the solver frequently produced numerical instabilities, including NaN values, preventing the optimization from progressing reliably.

4 Optimization Formulation

4.1 Design Variables

Aerodynamic and Planform Variables: The spanwise twist distribution of the main wing and horizontal tail, parametrized through B-spline control points (`twist_cp`), was selected as a design variable. By redistributing the local angle of attack along the span, twist allows the optimizer to shape the lift distribution towards a more elliptical form, reducing induced drag, while simultaneously shifting load inboard (washout) to relieve the bending moment at the wing and tail root. For the horizontal tail, the same control points additionally provide the rotational degree of freedom required for pitch trim, since a uniform shift of all tail twist control points is equivalent to a change in tail incidence; the optimizer therefore uses this single set of variables both to shape the tail's local lift distribution and to satisfy the cruise pitching-moment constraint. The thickness-to-chord ratio distribution, t/c, was also included, as it simultaneously affects profile drag and the depth, and therefore the bending stiffness and strength, of the wingbox cross-section. Furthermore, the spans of both the main wing (b_{wing}) and the horizontal tail (b_{tail}), as well as the wing sweep angle (Λ), were introduced as design variables. Allowing the span to vary enables the optimizer to explore the fundamental trade-off between aerodynamic efficiency (higher span reduces induced drag) and structural weight (higher span increases root bending moments). The sweep angle influences both the longitudinal aerodynamic center and the structural load paths, further expanding the design space and coupling the aerodynamic and structural sizing problems.

Structural Variables: The front and rear spar cap thickness distributions were chosen as design variables, allowing the optimizer to add material near the root, where bending moments are largest under the $n = 2.5$ maneuver, while thinning the spar towards the tip to minimize structural, and therefore empty, weight subject to the failure constraint. The same rationale applies to the upper and lower skin thickness distributions, which carry shear load and contribute to bending stiffness.

Operational Variables: The angle of attack at the cruise condition, α_c, and at the maneuver condition, α_m, were defined as independent design variables, allowing the optimizer to satisfy $L = W$ at cruise and $L = nW$ during the 2.5g pull-up, respectively.

4.2 Constraints

Cruise: The vertical-force equilibrium constraint, $L = W$, enforces that the trimmed lift generated by the wing-tail system exactly balances the aircraft weight at the cruise condition. The longitudinal trim constraint, $C_{M_y} = 0$, enforces that the total pitching moment about the center of gravity vanishes, ensuring the aircraft is statically trimmed in pitch at cruise; this is satisfied through the horizontal tail's twist/incidence degree of freedom.

Maneuver: The load factor constraint, $L = nW$ with $n = 2.5$, enforces the lift required to sustain the symmetric pull-up condition. The structural failure constraint, $\text{failure} \leq 0$, limits the Von Mises stress in the wing and tail spar caps and skins, evaluated through OpenAeroStruct's Kreisselmeier-Steinhauser (KS) aggregation of the individual stress ratios. Because the KS function is a smooth, conservative approximation of the true maximum stress ratio rather than an exact element-by-element check, satisfying $\text{failure} \leq 0$ guarantees the aggregated peak stress remains below yield under the 2.5g load, rather than a literal pointwise guarantee at every finite element.

Aerodynamic Constraints: To ensure that the flow remains attached and that OpenAeroStruct's Vortex Lattice Method (VLM) remains mathematically valid, the local sectional lift coefficients (C_l) were strictly constrained to the linear region of the lift curve. The maximum linear limits were predetermined using XFOIL by tracing the C_l versus α curves, present in Figures 9 (Appendix A.3), for the respective airfoils. Consequently, upper bounds were set at $C_l \leq 1.37$ for the main wing (NACA 43018) and $C_l \leq 1.19$ for the horizontal tail (NACA 0009) [1]. This actively prevents the optimizer from driving the aircraft into an unmodeled stall regime, particularly during the high angle of attack required for the 2.5g maneuver.

Geometric Constraints: A fuel volume constraint, $V_f \leq V_{wing}$, ensures the required fuel volume fits within the available wing tank volume.

Table 6: Summary of Aerostructural Optimization Parameters, Variables, and Constraints

Category	**Parameter**
	Design Variables
Aerodynamic & Planform	Wing & Tail Twist (θ_w, θ_t)
	Thickness-to-chord ratio (t/c)
	Wing & Tail Span (b_{wing}, b_{tail})
	Wing Sweep Angle (Λ)
Structural	Spar Cap Thickness (t_{spar})
	Skin Thickness (t_{skin})
Operational	Angle of Attack (α_c, α_m)
	Constraints
Cruise	Vertical Force Equilibrium ($L = W$)
	Pitching Moment Trim ($C_{M_y} = 0$)
Maneuver	Load Factor Lift ($L = 2.5W$)
	Structural Failure ($\text{failure} \leq 0$)
Aerodynamic	Sectional Lift Limit ($C_{l,\text{wing}} \leq 1.37$, $C_{l,\text{tail}} \leq 1.19$)
Fixed Baseline Parameters (ATR 72-600)	
Geometry	Root chord (c_{root}), taper (λ)
Environment	Mach number (M_{cruise}), Altitude (h)
Mass	Empty Weight + Payload (W_{fixed})
	Assumed Fuel Weight (W_f)
Material	Young's/Shear Modulus (E, G), Yield Stress (σ_{yield})
Propulsion	Specific Fuel Consumption (SFC)

5 Results and Discussion

This section presents the baseline (un-optimized) design, the build-up of the problem from a few to the complete set of design variables, the optimized configurations, and the analysis of the optimization history and cost.

5.1 Baseline Simulation

The baseline simulation serves as the starting point of the aerostructural optimization process, establishing the reference state of the ATR 72-600 prior to any geometric or structural modifications. The primary purpose of this phase is to evaluate the aerodynamic behavior and structural integrity of the aircraft under initial design conditions while simultaneously ensuring force and moment equilibrium (trimming).

5.1.1 Design Variables and Constraints

The optimization problem incorporates a minimal set of operational variables and physical boundaries, defined as follows:

Design Variables:

- Cruise Angle of Attack (α)
- Maneuver Angle of Attack ($\alpha_{maneuver}$)
- Horizontal Tail Twist (twist$_{tail}$)

Constraints:

- Vertical equilibrium ($L = W$) at $1.0g$ and $2.5g$
- Trim pitching moment ($C_M = 0$ in cruise)
- Wing failure index ($\gamma_{wing} \leq 0$) at $2.5g$
- Tail failure index ($\gamma_{tail} \leq 0$) at $2.5g$

5.1.2 Baseline Formulation

The mathematical architecture of the baseline optimization is structured to maximize the aircraft's range based on the Breguet equation:

$$\begin{aligned}
\text{Maximize} \quad & \text{Range} \\
\text{w.r.t.} \quad & \alpha,\ \alpha_{maneuver},\ \text{twist}_{tail} \\
\text{subject to} \quad & (L = W)_{1.0g} \\
& (L = W)_{2.5g} \\
& -10^{-4} \leq C_M \leq 10^{-4} \\
& \gamma_{wing,2.5g} \leq 0 \\
& \gamma_{tail,2.5g} \leq 0
\end{aligned} \tag{2}$$

5.1.3 Baseline Results

The parameters, objective values, and constraint states captured at the end of the execution are summarized in Table 7.

The baseline execution did not achieve mathematical convergence, terminating via Exit Mode 8 (*Positive directional derivative for linesearch*) after 8 iterations. This outcome is directly attributed to a critical structural violation: during the $2.5g$ symmetric maneuver, the wing failure index reached **0.8625**, indicating that von Mises stresses surpassed the material's yield strength by 86.25%.

With the skin and spar thicknesses locked as fixed parameters, the optimization algorithm lacked the geometric authority to reinforce the heavily loaded wing sections. To satisfy the mandatory $2.5g$ lift requirement, the wing undergoes severe bending moments that exceed its nominal capacity. Because this structural breakdown cannot be resolved by modifying purely aerodynamic or attitude states (α or tail twist), the line search procedure stalled. This non-converged state confirms that the baseline structure is insufficient for the full envelope, justifying the introduction of thickness distributions as active design variables in subsequent phases.

Table 7: Results and Optimization Convergence for the ATR 72-600 Baseline

Category	Parameter	Baseline Value
Variables	α [deg]	-0.0192
	$\alpha_{maneuver}$ [deg]	3.6094
	twist$_{tail}$ [deg]	0.0053
Objectives	Range [km]	1624.39
	Wingbox Mass [kg]	2531.74
Constraints	C_M	-1.00×10^{-4}
	$(L = W)_{1.0g}$	2.69×10^{-8}
	$(L = W)_{2.5g}$	-0.0062
	$\gamma_{wing,2.5g}$	**0.8625**
	$\gamma_{tail,2.5g}$	-0.5425
Optimization	Iterations	8
	Function Evaluations	14
	Gradient Evaluations	4
	Exit Mode	8 (FAILED)

5.2 Structural Optimization Phase 1A: Simple Structural Design

5.2.1 Objective of Phase 1A

The first structural optimization stage corresponds to the *Wing-Only Structural Optimization*, where only the structural design variables of the wing were allowed to vary, while the horizontal tail structural thicknesses were kept fixed. In addition to the wing structural variables, the trim variables required to satisfy the lift and pitching moment equilibrium were also included. The objective of this phase was the maximization of the Breguet range through the reduction of wing structural mass, while respecting all aerodynamic and structural constraints.

Compared to the baseline configuration, this stage introduces a moderately expanded design space, including:

- Wing spar thickness control points: `wing.spar_thickness_cp`
- Wing skin thickness control points: `wing.skin_thickness_cp`
- Cruise angle of attack: α
- Maneuver angle of attack: α_{man}
- Tail twist control point (for moment trim only): `tail.twist_cp`

5.2.2 Optimization Formulation

The optimization problem for Phase 1A was defined as a wing-only structural optimization, where only the wing structural thicknesses were allowed to vary, while the horizontal tail structural thicknesses remained fixed. The trim variables and the tail twist were included to ensure that the lift and pitching moment equilibrium constraints could be satisfied.

The optimization problem was formulated as:

$$\max R \quad \text{w.r.t. } \{\alpha,\ \alpha_{\text{man}},\ \text{tail.twist}_{cp},\ \text{wing.spar}_{cp},\ \text{wing.skin}_{cp}\}$$

subject to:

$$L = W,\ L_{\text{maneuver}} = 2.5\,W,\ CM = 0,\ \text{failure}_{\text{wing}} \leq 0,\ \text{failure}_{\text{tail}} \leq 0.$$

and the structural bounds:

$$0.002 \leq \text{wing thicknesses} \leq 0.05.$$

The tail structural thicknesses were intentionally kept fixed in this phase, as the objective of Phase 1A was to isolate the effect of wing structural optimization before introducing the full aero-structural coupling in Phase 1B.

5.2.3 Phase 1A Results

Table 8: Numerical performance of the SLSQP optimizer (Phase 1A).

Metric	Value
SLSQP iterations	21
Function evaluations	27
Gradient evaluations	21
Convergence status	Success (Exit mode 0)

Table 9: Optimized structural and trim results obtained in Phase 1A.

Quantity	Value
Optimized range $R^{\star}$	4390.97 km
Wingbox mass m_{wing}	1704.21 kg
Wing spar thickness	$[0.0020,\ 0.00293]$ m
Wing skin thickness	$[0.0020,\ 0.01695]$ m
Tail spar thickness	Fixed
Tail skin thickness	Fixed
Trim angle α	$3.73°$
Maneuver angle α_{man}	$8.64°$
Tail twist twist_{cp}	$-2.12°$
Wing failure (2.5g)	2.85×10^{-5}
Tail failure (2.5g)	-0.426
Pitching moment CM	1.0×10^{-4}

5.2.4 Phase 1A Conclusion

Phase 1A led to a cleaner redistribution of wing structural mass, since only the wing thicknesses were allowed to vary. The optimizer slightly reinforced the wing tip region, while keeping the root at the minimum allowable thickness, consistent with the load distribution predicted by OpenAeroStruct.

Because the tail structure remained fixed, the tail failure constraint stayed well below its limit, whereas the wing failure converged close to zero, as expected. The trim variables α, α_{man}, and the tail twist adjusted automatically to satisfy lift and moment equilibrium.

Overall, Phase 1A produced a structurally efficient wing operating near its allowable stress limits, serving as a controlled intermediate step before the full optimization in Phase 1B.

5.3 Structural Optimization Phase 1B: Full Structural Design

5.3.1 Objective of Phase 1B

The second optimization stage corresponds to the *Full Structural Optimization*, where all structural design variables of both the wing and the horizontal tail were allowed to vary, together with the trim variables required to maintain longitudinal equilibrium. The objective remained the maximization of the Breguet range through the reduction of structural mass, while satisfying all aerodynamic and structural constraints.

Compared to Phase 1A, this stage introduces a significantly larger design space, including:

- Wing spar thickness control points: `wing.spar_thickness_cp`
- Wing skin thickness control points: `wing.skin_thickness_cp`
- Tail spar thickness control points: `tail.spar_thickness_cp`
- Tail skin thickness control points: `tail.skin_thickness_cp`
- Cruise angle of attack: α
- Maneuver angle of attack: α_{man}
- Tail twist control point: `tail.twist_cp`

5.3.2 Optimization Formulation

The optimization problem was formulated as follows:

$$\max R \quad \text{w.r.t. } \{\alpha,\ \alpha_{\text{man}},\ \text{tail.twist}_{cp},\ \text{wing.spar}_{cp},\ \text{wing.skin}_{cp},\ \text{tail.spar}_{cp},\ \text{tail.skin}_{cp}\}$$

s.t. $L = W,\ L_{\text{maneuver}} = 2.5\,W,\ CM = 0,\ \text{failure}_{\text{wing}} \le 0,\ \text{failure}_{\text{tail}} \le 0,\ 0.002 \le \text{wing thk} \le 0.05,\ 0.001 \le \text{tail thk} \le 0.05.$

Solver Settings The SLSQP optimizer was used with tolerance 10^{-3} and a maximum of 80 iterations.

Table 10: Numerical performance of the SLSQP optimizer (Phase 1B).

Metric	**Value**
SLSQP iterations	24
Function evaluations	28
Gradient evaluations	24
Convergence status	Success (Exit mode 0)

Table 11: Optimized structural results obtained in Phase 1B.

Quantity	**Value**
Optimized range $R^{\star}$	4298.59 km
Wingbox mass m_{wing}	1665.86 kg
Wing spar thickness	$[0.0020,\ 0.00299]$ m
Wing skin thickness	$[0.0020,\ 0.01650]$ m
Tail spar thickness	$[0.0010,\ 0.0010]$ m
Tail skin thickness	$[0.0010,\ 0.00172]$ m
Wing failure (2.5g)	1.29×10^{-8}
Tail failure (2.5g)	3.73×10^{-7}
Pitching moment CM	9.99998×10^{-5}

5.3.3 Phase 1B Results

5.3.4 Phase 1B Conclusion

Allowing all wing and tail structural variables to vary enabled a more efficient redistribution of structural mass. The optimizer drove the tail thicknesses to their minimum bounds, while the wing required reinforcement near the tip, consistent with the higher loads predicted by OpenAeroStruct.

The trim variables α, α_{man}, and the tail twist adjusted automatically to satisfy lift and moment equilibrium in both flight conditions.

Overall, Phase 1B produced a fully optimized aero-structural configuration operating at the failure limits, as expected in a well-posed structural optimization.

5.4 Aerodynamic Optimization

Following the structural optimization phase, this stage focuses on maximizing the aircraft's performance by modifying its external aerodynamic shape and planform parameters. While the internal structural thicknesses remain fixed at this step, the geometric envelope is unlocked to minimize aerodynamic drag and enhance lifting efficiency, directly translating to an extension of the aircraft's operational range.

5.4.1 Optimization Parameters and Bounds

To prevent unrealistic geometries while maximizing aerodynamic efficiency, a select group of five geometric parameters was chosen alongside the attitude variables. The design space and boundary limits established for this problem are defined below:

Design Variables:

- Operational Trim: α, $\alpha_{maneuver}$
- Wing: b_{wing}, Λ_{wing}, θ_{wing}
- Tail: b_{tail}, θ_{tail}

Constraints:

- Lift-to-weight equilibrium ($L = W$) at $1.0g$ and $2.5g$
- Longitudinal trim ($C_M = 0$)
- Yield boundaries ($\gamma \leq 0$) at $2.5g$

5.4.2 Mathematical Problem Formulation

The mathematical architecture is formulated to directly maximize the Breguet flight range (R) under strict enforcement of the aeroelastic trim and structural safety constraints:

$$\begin{aligned}
\text{Maximize} \quad & R \\
\text{w.r.t.} \quad & \alpha,\ \alpha_{maneuver},\ \theta_{wing},\ \theta_{tail},\ b_{wing},\ b_{tail},\ \Lambda_{wing} \\
\text{subject to} \quad & (L = W)_{1.0g} \\
& (L = W)_{2.5g} \\
& -10^{-4} \leq C_M \leq 10^{-4} \\
& \gamma_{wing,2.5g} \leq 0 \\
& \gamma_{tail,2.5g} \leq 0
\end{aligned} \tag{3}$$

5.4.3 Quantitative Results and Physical Discussion

The numerical problem was solved using the gradient-based `SLSQP` algorithm. In contrast to the baseline simulation, the optimizer achieved full mathematical convergence (*Exit Mode 0*) in 19 iterations, involving 33 function evaluations. The comprehensive telemetry from the converged state is compiled in Table 12.

Table 12: Results and Optimization Convergence for the Aerodynamic Phase

Category	Parameter	Optimized Value
Variables	α [deg]	-0.0192
	$\alpha_{maneuver}$ [deg]	3.6094
	Wing Span (b_{wing}) [m]	**28.50**
	Tail Span (b_{tail}) [m]	**7.00**
	Wing Sweep (Λ_{wing}) [deg]	**0.00**
	Wing Twist (θ_{wing}) [deg]	$[-10.00,\ -9.93,\ 3.59]$
	Tail Twist (θ_{tail}) [deg]	-3.47
Objectives	Range [km]	3625.92
	Wingbox Mass [kg]	1479.83
Constraints	C_M	$\mathbf{-1.01 \times 10^{-4}}$
	$(L = W)_{1.0g}$	2.69×10^{-8}
	$(L = W)_{2.5g}$	-0.0062
	$\gamma_{wing,2.5g}$	$\mathbf{-2.11 \times 10^{-6}}$
	$\gamma_{tail,2.5g}$	-0.7921
Optimization	Iterations	19
	Function Evaluations	33
	Gradient Evaluations	19
	Exit Mode	0 (Success)

The physical trends driven by the optimizer reveal a highly logical behavior for a regional turboprop configuration:

- **Wing and Tail Spans:** The wing span (b_{wing}) expanded to its absolute upper limit of 28.5 m. This maximizes the aspect ratio, minimizing induced drag (D_i) and directly inflating the flight range to 3625.92 km. Conversely, the horizontal tail span (b_{tail}) contracted to its lower bound of 7.0 m to minimize parasite skin-friction drag and component weight, retaining just enough aerodynamic authority to balance the pitch axis.

- **Wing Sweep:** The sweep angle (Λ_{wing}) converged to $0°$. For an aircraft operating at a moderate Mach number ($M = 0.443$), compressibility effects are non-existent; thus, the optimizer removes any sweep to maximize effective lift and avoid structural weight penalties.

- **Twist and Structural Feasibility:** The wing twist control points adapted into a prominent washout distribution, with the inboard sections hitting the lower bound of $-10°$. This aggressive twist unloads the wingtips, shifting the aerodynamic load center inboard. This spanwise lift redistribution successfully mitigates the root bending moments during the $2.5g$ pull-up, driving the wing failure index precisely to zero ($\gamma_{wing} \approx 0$) and rendering the final design structurally viable without requiring a change in material thickness.

5.5 Aerostructural Optimization

The final phase involves a fully coupled aerostructural optimization, where both external aerodynamic shape and internal structural dimensions are resolved simultaneously. By allowing the optimizer to exploit the multi-disciplinary synergies between aerodynamic lift distribution and structural wingbox mass, the algorithm can navigate trade-offs that separate discipline evaluations cannot capture, aiming for an absolute maximization of the Breguet flight range.

5.5.1 Design Variables and Constraints

To handle the increased complexity of the coupled problem, the entire design space is unlocked simultaneously. The parameters and boundaries are structured side-by-side as follows:

Design Variables:

- **Aerodynamic/Trim:** α, $\alpha_{maneuver}$
- **Global Shape:** b_{wing}, b_{tail}, Λ_{wing}
- **Local Twists:** θ_{wing} (3-cp), θ_{tail} (1-cp)
- **Wing Structure:** $t_{spar,wing}$, $t_{skin,wing}$ (2-cp)
- **Tail Structure:** $t_{spar,tail}$, $t_{skin,tail}$ (2-cp)

Constraints:

- Vertical equilibrium ($L = W$) at $1.0g$ and $2.5g$ (rigid unit scaler)
- Cruise pitching moment status ($-10^{-3} \leq C_M \leq 10^{-3}$)
- Wing yield structural boundary ($\gamma_{wing,2.5g} \leq 0$)
- Tail yield structural boundary ($\gamma_{tail,2.5g} \leq 0$)

5.5.2 Problem Formulation

The complete multi-disciplinary optimization problem is mathematically defined to maximize the operational range, fully driving the coupling between structural weight and aerodynamic coefficients:

$$
\begin{aligned}
\text{Maximize} \quad & R \\
\text{w.r.t.} \quad & \alpha,\ \alpha_{maneuver},\ b_{wing},\ b_{tail},\ \Lambda_{wing},\ \theta_{wing},\ \theta_{tail},\ t_{spar},\ t_{skin} \\
\text{subject to} \quad & (L = W)_{1.0g} = 0 \\
& (L = W)_{2.5g} = 0 \\
& -10^{-3} \leq C_M \leq 10^{-3} \\
& \gamma_{wing,2.5g} \leq 0 \\
& \gamma_{tail,2.5g} \leq 0
\end{aligned}
\tag{4}
$$

5.5.3 Results and Convergence

The coupled problem successfully converged to an optimal solution (*Exit Mode 0*) after 31 iterations and 60 function evaluations. The final telemetry for the optimized ATR 72-600 is detailed in Table 13.

Table 13: Results and Optimization Convergence for the Aerostructural Phase

Category	Parameter	Optimized Value
Variables	α [deg]	5.12
	$\alpha_{maneuver}$ [deg]	11.65
	Wing Span (b_{wing}) [m]	**20.00**
	Tail Span (b_{tail}) [m]	**7.00**
	Wing Sweep (Λ_{wing}) [deg]	5.48
	Wing Twist (θ_{wing}) [deg]	$[-0.34,\ 1.70,\ 0.36]$
	Tail Twist (θ_{tail}) [deg]	-3.26
	Wing Spar Thickness ($t_{spar,w}$) [mm]	$[\mathbf{5.00},\ \mathbf{5.00}]$
	Wing Skin Thickness ($t_{skin,w}$) [mm]	$[5.00,\ 12.06]$
	Tail Spar Thickness ($t_{spar,t}$) [mm]	$[\mathbf{2.00},\ \mathbf{2.00}]$
	Tail Skin Thickness ($t_{skin,t}$) [mm]	$[\mathbf{2.00},\ \mathbf{2.00}]$
Objectives	Range [km]	4911.54
	Wingbox Mass [kg]	1171.03
Constraints	C_M	$\mathbf{-1.00 \times 10^{-3}}$
	$(L = W)_{1.0g}$	4.98×10^{-7}
	$(L = W)_{2.5g}$	6.80×10^{-7}
	$\gamma_{wing,2.5g}$	$\mathbf{2.95 \times 10^{-6}}$
	$\gamma_{tail,2.5g}$	-0.3397
Optimization	Iterations	31
	Function Evaluations	60
	Gradient Evaluations	30
	Exit Mode	0 (Success)

The aerostructural synthesis produced an optimized flight range of **4911.54 km**, a substantial improvement over the previous stages driven by a classic multidisciplinary trade-off:

- **The Span-Weight Trade-off:** In stark contrast to the pure aerodynamic optimization, the wing span (b_{wing}) contracted to its absolute lower bound of **20.00 m**. While a smaller span increases induced drag, it drastically curtails the aerodynamic bending moments acting on the wing root.

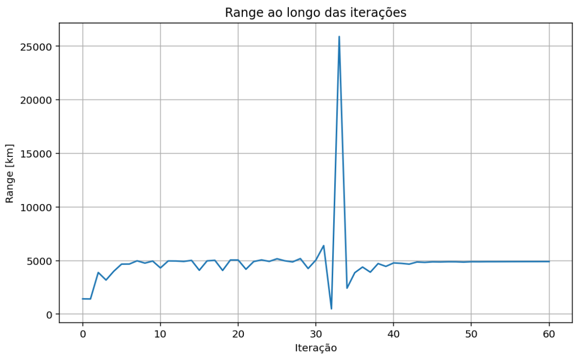
This allowed the optimizer to slash the wingbox structural mass down to an exceptionally low value of **1171.03 kg** (compared to 1479.83 kg in the aerodynamic phase). According to the Breguet range equation, this structural mass reduction heavily dominates the logarithmic mass ratio term, yielding a net gain in range that overcomes the aerodynamic drag penalty.

- **Aerodynamic Compensation:** To compensate for the loss in lifting area caused by the shorter span, the aircraft is forced to fly at a significantly higher cruise angle of attack ($\alpha = 5.12°$), pushing the cruise lift coefficient to a high nominal value ($C_L = 0.651$).
- **Active Constraints and Sizing:** The structural optimization shows that the wing spar thickness and the outboard wing skin hit their minimum manufacturing bounds of **5.00 mm**. To safely sustain the $2.5g$ pull-up maneuver at an aggressive angle of attack ($\alpha_{maneuver} = 11.65°$), the optimizer locally reinforced the inboard wing skin thickness to **12.06 mm**. This precise structural tailoring drove the wing failure index exactly to its mathematical limit ($\gamma_{wing} \approx 0$), proving that the wing is fully optimized and structurally critical. Concurrently, the longitudinal trim constraint (C_M) hit its lower allowable bound of -1.00×10^{-3}, meaning the horizontal tail is operating at its maximum balancing capacity.

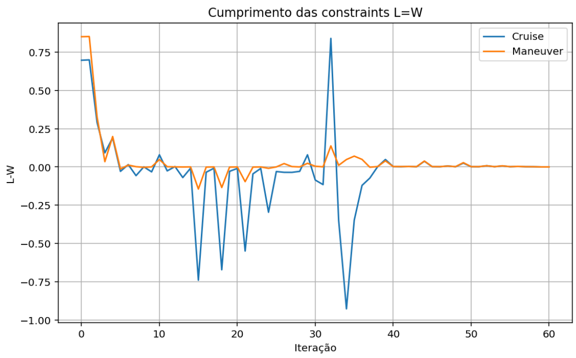
5.6 Aerostructural Optimization History

The aero-structural optimization history is included to illustrate how the coupled aerodynamic and structural variables evolve throughout the design process. These plots provide essential insight into the solver's convergence behavior and highlight the interactions that drive the final optimized configuration.

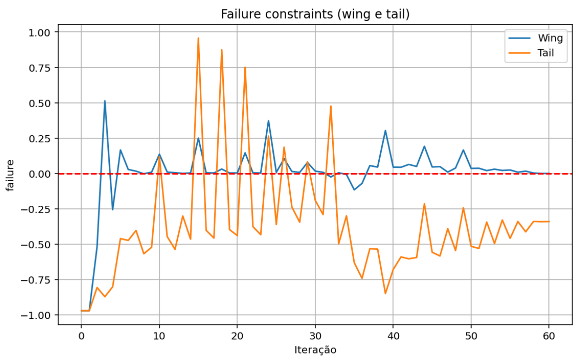
Constraints history



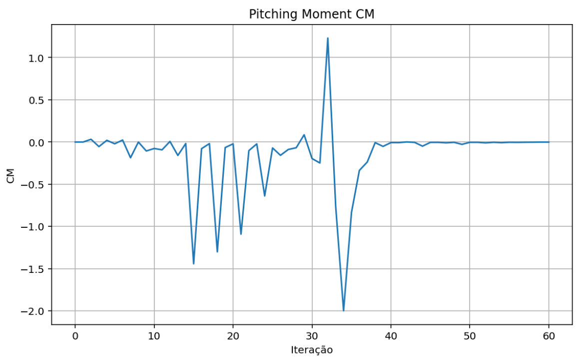
Range over iterations



Lift equals Weight constraint



Structural failure constraints



Pitching moment coefficient

Figure 7: Combined visualization of the four main optimization history plots: range evolution, lift–weight constraint satisfaction, structural failure constraints, and pitching moment convergence.

Figure 7 shows that regarding:

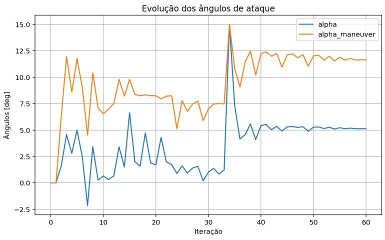
- The range remains nearly constant throughout the optimization, indicating stable aerodynamic and structural performance. A single abrupt spike appears mid-process, showing that the solver briefly explored an infeasible or highly non-optimal region before returning to a consistent solution.shows that the predicted range remains essentially stable during the optimization, except for a single abrupt spike caused by the solver exploring an infeasible region before converging back to a physically consistent solution.

- Both the cruise and maneuver conditions progressively converge toward zero, showing that the optimizer successfully adjusts the aerodynamic variables to satisfy the lift-equals-weight constraint. The oscillations in the early iterations reflect the solver's search for a feasible trim configuration before stabilizing near the constraint boundary.

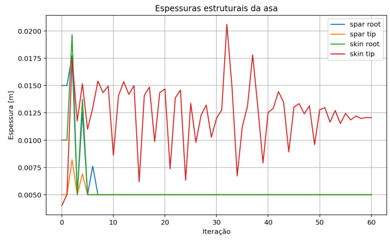
- Both the wing and tail failure constraints oscillate around the zero threshold, showing that the optimizer actively adjusts structural thicknesses to satisfy the strength requirements. The gradual reduction of excursions above zero indicates that the design becomes progressively safer and converges toward a structurally feasible solution

- The pitching moment coefficient exhibits large oscillations during the early iterations, reflecting the optimizer's effort to balance the aerodynamic moments through twist and angle-of-attack adjustments. As the optimization progresses, these fluctuations diminish and CM converges toward zero, indicating that a trimmed and longitudinally stable configuration has been achieved.

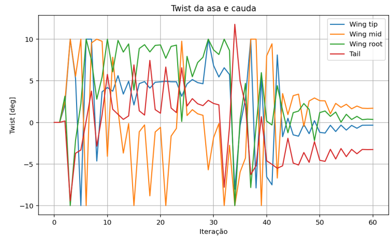
Variables history



Angle of attack variables



Structural thickness evolution



Wing and tail twist control points

Figure 8: Evolution of the main design variables: angle of attack, structural thickness, and twist distribution.

Regarding the variables behaviour throughout the optimization:

- The angle of attack variables show a clear convergence trend, with both α and $\alpha_{maneuver}$ stabilizing after an initial phase of oscillations. The higher variability of the maneuver angle reflects the optimizer's adjustments to meet more demanding aerodynamic conditions before settling into a consistent trim state.
- The structural thickness variables show distinct convergence behaviors, with the spar thicknesses stabilizing smoothly while the skin tip thickness undergoes larger oscillations due to localized load-driven adjustments. The constant value at the skin root indicates that this region quickly satisfies the structural constraints and remains unchanged throughout the optimization.
- The twist variables show significant adjustments across the wing and tail, reflecting the optimizer's effort to redistribute aerodynamic loading along the span. As the iterations progress, the twist values gradually settle, indicating convergence toward a configuration that balances lift distribution and pitching-moment requirements.

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6 Conclusions

This study presented a comprehensive aerostructural optimization of the ATR 72-600 using the OpenAeroStruct framework, integrating aerodynamic performance, structural integrity, and trim requirements into a unified multidisciplinary formulation. The development process followed a progressive approach, starting from baseline flight conditions and evolving into fully coupled aeroelastic analyses under cruise and high-load maneuver scenarios.

The results demonstrate the efficacy of a consistent Multidisciplinary Design Optimization (MDO) approach. The optimizer successfully coordinated geometric design variables, such as twist distribution, structural thickness, and span, to satisfy lift and trim requirements while significantly improving the aircraft's range relative to the baseline. The optimization histories reveal complex interactions between aerodynamic loading and structural response, illustrating how stiffness and twist distributions are inherently coupled to maintain equilibrium under load.

Naturally, the study acknowledges the limitations inherent to low-fidelity tools such as OpenAeroStruct. Simplified VLM aerodynamics, the exclusion of fuselage and propulsion modeling, and the idealized wingbox representation introduce inevitable deviations from real-world aircraft behavior. Furthermore, numerical stability in advanced optimization phases demanded rigorous tuning of discretization and solver parameters. Despite these simplifications, the observed trade-offs remain physically consistent, providing meaningful insights into aerostructural design.

Overall, this project provided a valuable hands-on experience in conceptual aircraft design, reinforcing the necessity of coupling aerodynamic and structural effects. The methodology developed establishes a solid foundation for future work, potentially using higher-fidelity tools, and confirms the potential of MDO frameworks to support efficient, integrated design decisions in early-stage aeronautical engineering.

A Appendix

A.1 ATR 72-600 reference data

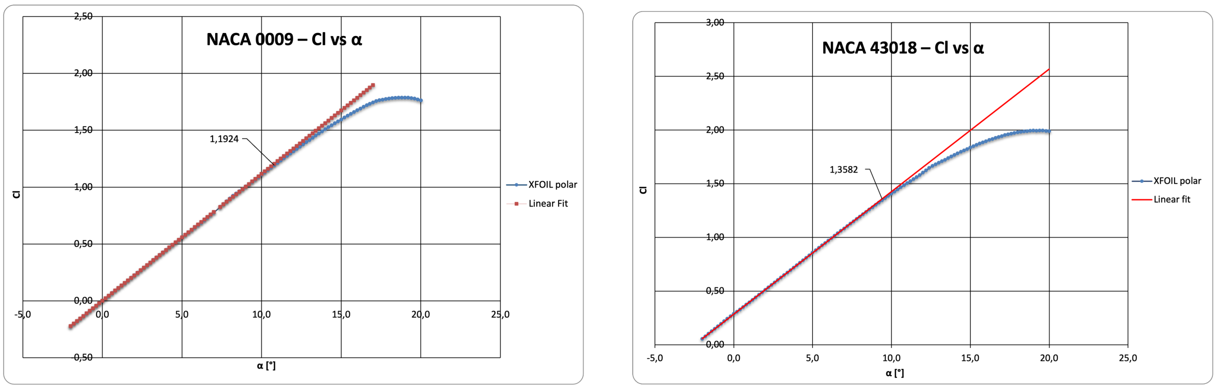
Table 14 consolidates every input value used to build the model, together with the source of each one. The geometry, weights, range and engine come directly from the manufacturer factsheet; the wing aerodynamic parameters from the Nita design study; the remaining values are either derived from these (ISA atmosphere, geometric relations) or are standard material/assumption values.

Parameter	Value	Parameter	Value
Cruise speed (m/s)[c]	137.2	Main wing span (m)[a]	27.05
Cruise Mach[b]	0.443	Main wing area (m^2)[a]	61.0
Cruise altitude (m)[a]	7620	Main wing aspect ratio[a]	12.0
Range (km)[a]	1404	Root / tip chord (m)[c]	3.11 / 1.40
Fuel consumption (kg/h)[a]	~760	Mean aerodynamic chord (m)[c]	2.36
Fuel density (kg/m^3)[d]	803	Wing airfoil (root/tip)[b]	NACA 43018/43013
Operating empty weight (kg)[a]	13 450	Wing t/c (root/tip)[b]	0.18 / 0.13
Take-off weight (kg)[a]	23 000	Wing max-thickness loc.[b]	0.30 c
Equiv. TSFC C_T (1/s)[d]	1.25e-4	Wing max C_L (clean)[b]	1.374
Engine weight (kg)[a]	480	Tail area (m^2)[c]	12.0
Cruise air density (kg/m^3)[c]	0.549	Tail span (m)[a]	7.43
Cruise speed of sound (m/s)[c]	309.6	Tail aspect ratio[a]	4.6
Cruise Reynolds/chord (1/m)[c]	4.9e6	Tail taper ratio[c]	0.5
Load factor n[d]	2.5	Tail airfoil[b]	NACA 0009
Wing sweep @$c/4$ (°)[b]	3	Tail t/c[b]	0.09
Wing taper ratio[b]	0.45	Structural material[d]	Al 7075-T6
Wing dihedral (°)[b]	0	Young modulus E (Pa)[d]	71.7e9
Tail moment arm (m)[c]	12.0	Shear modulus G (Pa)[d]	26.9e9
Maneuver altitude (m)[d]	0 (SL)	Yield stress (Pa)[d]	505e6
		Material density (kg/m^3)[d]	2810

Table 14: Complete set of input data used to build the ATR 72-600 model. [a] ATR 72-600 manufacturer factsheet [2]; [b] Nita ATR 72 design study [7]; [c] derived from other parameters (ISA atmosphere, geometric relations, tail-volume method); [d] standard value or model assumption (n per EASA CS-25 [**CS25**]; Al 7075-T6 properties from material tables; equivalent TSFC calibrated to the published cruise fuel flow).

A.2 Curve of C_l vs. α

The lift polars of the NACA 43018 (wing) and NACA 0009 (tail) airfoils were obtained in XFOIL at $Re = 11.6 \times 10^6$ and $Re = 8.2 \times 10^6$, respectively, computed from the cruise conditions at FL250 ($\rho = 0.549$ kg/m^3, $V = 136.9$ m/s) using the mean chord of each surface. The adopted C_l limit is not the absolute maximum of the polar, but the point at which the actual curve begins to diverge from the linear fit: $C_l = 1.37$ for the NACA 43018 and $C_l = 1.19$ for the NACA 0009. This choice follows from the VLM assuming linear potential flow, which is physically valid only within that region. The constraints $C_l \leq 1.37$ and $C_l \leq 1.19$ therefore ensure that the optimiser does not venture beyond the validity domain of the aerodynamic model.



(a) NACA 0009

(b) NACA 43018

Figure 9: C_l vs. α for the two airfoils used.

A.3 Use of AI

All decisions and technical content of this work were authored by Afonso Vieira, Gonçalo Teixeira, Guilherme Rodrigues, and Cândida Miranda. The following AI tool was used: Gemini of Google (accessed in June 2026) [https://gemini.google.com/] for linguistic correction, text restructuring, mathematical consistency proofreading, and support in the construction and debugging of the computational code.

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